Fourth-harmonic generation in a single lithium niobate-crystal with cascaded second-harmonic generation

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Tunable second- and fourth-harmonic radiation was generated in a single 1-cm-long lithium niobate ($LiNbO_3$) crystal with the Mark III infrared free electron laser at Duke University. The fundamental laser radiation was tuned from 2 to 2.5 μm, yielding 1–1.25-μm radiation (second harmonic) and 0.5–0.625-μm radiation (fourth harmonic). A fundamental–second-harmonic energy conversion efficiency of 66% and a fundamental–fourth-harmonic energy conversion efficiency of 3.3×10^{-6} were measured. The maximum energy in the fourth harmonic was 3.3 nJ.

Key words: Fourth-harmonic generation, cascaded second-harmonic generation, third-harmonic generation, $LiNbO_3$, free-electron laser.

Introduction

It is well known that the efficiency of parametric wave-mixing interactions, such as second-harmonic generation (SHG), is enhanced significantly when momentum is conserved in the interaction (phase-matched interaction). Because all nonlinear optical materials display dispersion in the refractive index, it is usually possible to phase match only a single (if any) wave-mixing process out of the large number of potential interactions that could occur in the nonlinear medium. This fact tends to increase the complexity of optimizing high-order interactions, such as a fourth-order interaction. An effective fourth-order nonlinearity has been developed with fourth-harmonic generation (FHG) in a single lithium formiate crystal.[1] The lithium formiate crystal was phase matched for direct FHG, and the influence of cascaded processes, including second-order and third-order nonlinearities, was determined. We have found that it is possible to obtain modest fourth-harmonic (FH) conversion efficiencies in a single crystal when it is phase matched for SHG.

In this paper we describe cascaded second-harmonic generation, SHG:SHG, as observed in a single 1-cm-long $LiNbO_3$ crystal, type I phase matched for the first-stage SHG. We denote the second-stage SHG as FHG for brevity. This SHG:SHG setup can be used as a source of multiphoton and pump-probe experiments. The three-color output of this setup is colinear, wavelength tunable, and synchronous in time, making it a very simple and desirable source for these types of experiments.

The Mark III infrared free electron laser (FEL) at Duke University[2] produces megawatts of peak power in short picosecond pulses, making it an ideal source for the investigation of nonlinear optical interactions in matter. The Mark III FEL is driven by a radio-frequency linear accelerator and operates in the 2–9-μm region of the infrared. Harmonic-generation experiments with the Mark III FEL have been carried out since 1987.[3–5]

FHG through SHG:SHG is typically performed with two nonlinear crystals, where each crystal is optimized to double the frequency of the input light efficiently. We have generated the FH, tunable from 490 to 630 nm, with a type I cascaded crystal system of $LiNbO_3$ and β barium borate.[4] Cascade interresonator generation of the FH of a neodymium-activated glass laser was reported in 1973.[6] More recently two external enhancement cavities[7] and a single ring cavity[8] were used for FHG of a Nd:YLF laser. These methods all use two crystals. Another scheme, known as quadrature frequency conversion,[9] uses two crystals for each step of frequency conversion, thus four crystals are needed for FHG. In the quadrature scheme, the *kz* planes in the two crystals

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are orthogonal to each other, and type II phase matching is used. This scheme has the advantage that the dynamic range, the range of intensity over which the efficiency is high, can be 10 to 100 times larger than that for a single crystal but with the added complexity of four crystals.

Experimental Setup and Results

A 1-cm-long $LiNbO_3$ crystal was cut to phase match SHG for wavelengths between 2 and 2.5 μm. The $LiNbO_3$ crystal is optimized for type I SHG but not for FHG. It was therefore surprising when we noticed visible light exiting the crystal. This visible light was the generated FH.

The FEL was tuned to a particular wavelength at which the pulse widths, energy, and spectral linewidth were measured. The full-width at half-maximum (FWHM) spectral linewidth $\Delta\lambda/\lambda$ was typically of the order of 0.5–1% for these experiments. The Mark III FEL has a micropulse train enclosed in a macropulse envelope. The width of the macropulses is typically 1–3 μs FWHM at a repetition rate of 10 Hz. The micropulse train is sent into a nonlinear optical autocorrelator to determine the second-order intensity autocorrelation function. The FWHM micropulse widths have typically ranged from 1 to 3 ps, at a repetition rate of 2.857 GHz as set by the electron accelerator, and the micropulse shape, inferred from the shape of the autocorrelation function, is that of a top hat.

Fast gold-doped germanium (Ge:Au) detectors (bandwidth greater than 150 MHz) and slow broadband pyroelectric detectors were used to measure simultaneously the intensity of the fundamental and the second-harmonic (SH) beams. A Germanium plate, set near Brewster's angle, was used to pick off a small percentage (1%) of the fundamental beam for characterization. In addition to the fundamental laser light, the bunched electrons in the undulator produce coherent spontaneous harmonics. Some of these higher harmonics are in the visible and are ideal for aligning the FEL light into a beam line, but for our purposes they would be confusing. Germanium, with a cutoff wavelength of 1.8 μm, was used because it effectively filters all but the fundamental wavelength. Because the SH is polarized 90° to the fundamental, a rotatable polarizer was used in tandem with filters to select a given polarization (harmonic) for measurement. A fast silicon photodiode, comparable to speed to the Ge:Au detectors, was used to measure the FH signal.

Measurements of the transverse beam profile indicate that the beam is Gaussian. The area of our Gaussian beam is therefore calculated as $(\pi\omega_0{}^2/2)$, where ω_0 is the $1/e^2$ beam radius. The measured fundamental FWHM pulse widths for these experiments were 1.8 μs for the macropulse and 2 ps for the micropulse. The micropulse widths for the SH and the FH were not measured. The FH was first noticed when the laser beam was focused to a 45-μm waist in the crystal. In this case, energy conversion efficiencies of 66% to the SH and 3.3×10^{-6} to the SH were measured for an input macropulse energy of 1 mJ, input intensity $[(1\ \text{mJ}/1.8\ \mu\text{s})/2.857\ \text{GHz}/2\ \text{ps}]/(\pi\omega_0{}^2/2) = 3\ \text{GW/cm}^2$, at a fundamental wavelength of 2 μm. Interference fringes resulting from the strong focusing were visible in the FH, similar to those described by Kleinman *et al.*,[10] although their study was concerned with SHG, not FHG. The laser beam was also focused to a 230-μm waist. We set the intensity to the same value by increasing the input power of the laser, and similar conversion efficiencies were measured for this configuration. However, the interference fringes were not observed for this weakly focused case. A fundamental–FH conversion efficiency of 5×10^{-7} was measured when the SHG conversion efficiency was as low as 25% and the fundamental input energy was approximately 167 μJ (0.5 GW/cm²). No FH was detectable when the input energy was below approximately 16.7 μJ (50 MW/cm²). Similar conversion efficiencies were obtained for input wavelengths from 2 to 2.5 μm. The polarization of the FH is the same as the polarization of the SH, and the FH signal is optimized when the SH signal is optimized. No third-harmonic was observed in these experiments.

Analysis of the Coupled Equations

The first query we made was to determine which interaction was responsible for the generation of the FH. Did the fundamental generate the FH by direct FHG? Or did the SH generate the FH by a cascaded process, SHG:SHG? To determine this we investigated solutions to the coupled nonlinear equations that describe the propagation of the three fields in the nonlinear crystal.

Consider a monochromatic plane wave propagating in the z direction. The coupled nonlinear equations describing the evolution of the slowly varying envelope for the fundamental, SH, and FH electric fields are given by

$$\frac{\partial E_1}{\partial z} = i\kappa_1 E_1{}^* E_2 \exp(i\Delta k_{2\omega} z) + i\kappa_4 E_1{}^{*3} E_4 \exp(i\Delta k_{\omega} z), \tag{1a}$$

$$\frac{\partial E_2}{\partial z} = i\kappa_1 E_1{}^2 \exp(-i\Delta k_{2\omega} z) + i\kappa_2 E_2{}^* E_4 \exp(i\Delta k_{4\omega} z), \tag{1b}$$

$$\frac{\partial E_4}{\partial z} = i\kappa_2 E_2{}^2 \exp(-i\Delta k_{4\omega} z) + i\kappa_4 E_1{}^4 \exp(-i\Delta k_{\omega} z), \tag{1c}$$

respectively, where $\kappa_i = \omega_1 d_{\text{eff}}/n_i c$ $(i = 1, 2, 4)$ are coupling constants. Other quantities are given by d_{eff}, the effective nonlinear optical coefficient; n, the index of refraction; c, the speed of light; ω, the angular frequency; $\Delta k_{\omega} = k_4 - 4k_1$, the phase velocity mismatch (phase mismatch) between the fundamental and the FH; $\Delta k_{2\omega} = k_2 - 2k_1$, the phase mismatch between the fundamental and the SH; and $\Delta k_{4\omega} = k_4 - 2k_2$, the phase mismatch between the SH and the FH. The coupling between the fundamental and the SH fields, κ_1, and between the SH and the FH fields, κ_2, is a $\chi^{(2)}$ process, whereas the coupling between the fundamental and the FH, κ_4, is a $\chi^{(4)}$ process, which is proportional to $[\chi^{(2)}]^2$.

The important parameters for determining the dominant interaction for generating the FH in Eqs. (1) are the coupling constants κ_2 and κ_4 and the phase mismatches Δk_ω and $\Delta k_{4\omega}$. We must compare the order of magnitude of these parameters for their combined effect on the coupled equations. First we compare the phase mismatch for each interaction when the crystal is phase matched for type I SHG ($\theta = \theta_m$) in $LiNbO_3$ at 2.2 μm. We need only be concerned with the two FHG phase-matching schemes, type I [ordinary (o) ray + o ray → [extraordinary (e) ray] and (e ray + e ray → e ray), for a negative uniaxial crystal. Figure 1(a) shows the phase mismatch of the two possible FHG interactions, Δk_ω(oe) (direct fundamental–FHG, filled circles) and $\Delta k_{4\omega}$(ee) (SHG:SHG or cascaded SHG, solid line), versus angle for phase-matched SHG in $LiNbO_3$ at 2.2 μm. Note that the phase mismatches for cascaded SHG and direct FHG are equal when the interaction is phase matched for SHG ($\theta = \theta_m$); i.e., $\Delta k_\omega = \Delta k_{4\omega}$ when $\Delta k_{2\omega} = 0$. Second we compare the magnitudes of the coupling constants. The direct generation of the FH is coupled by κ_4, which is proportional to the square of κ_1 and is therefore small. The FH field in Eq. (1c), however, is not

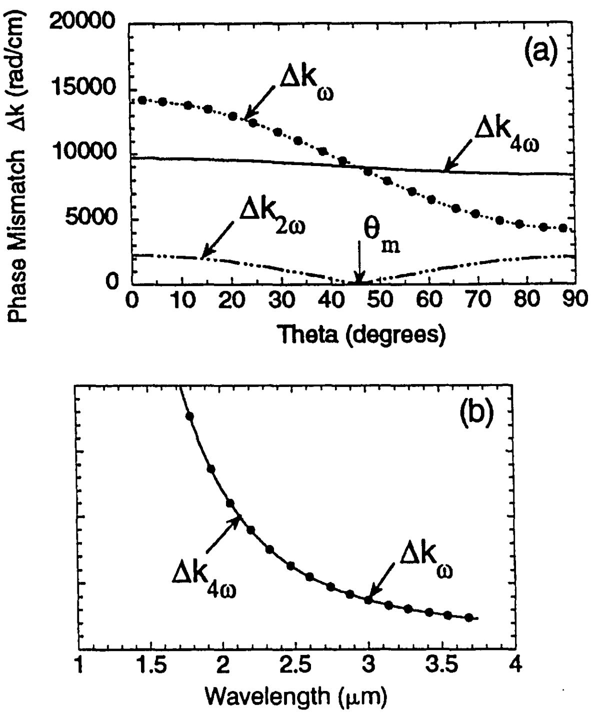

Fig. 1. Phase mismatch versus angle for FHG by cascaded SHG $\Delta k_{4\omega}$ (——) and by direct FHG Δk_ω when (a) type I phase matched for SHG ($\theta = \theta_m$) in $LiNbO_3$ at (a) λ = 2.2 μm and (b) type I phase matched for SHG in $LiNbO_3$, at each wavelength shown.

proportional to the fundamental field to the fourth power, because the fundamental field is depleted by the SHG interaction. The small coupling and large phase mismatch make direct interaction less probable.

Therefore it appears that the fundamental propagates as an o ray, and then generates the SH as an e ray, and that the e ray then generates the FH as another e ray. Figure 1(b) is a graph of the phase mismatch versus wavelength in $LiNbO_3$ for the two FHG interaction schemes when they are phase matched for SHG at each wavelength. Based on Fig. 1(b), because the phase mismatch is smaller at longer wavelengths, one would expect better conversion to the FH at longer wavelengths. But the propagation direction in the crystal and the magnitude of the coupling constants are also important.

We now calculate the phase mismatches $\Delta k_{2\omega}$ and $\Delta k_{4\omega}$ and the coupling constants κ_1 and κ_2 for the SHG:SHG coupled Eqs. (1) without the κ_4 terms. The coupling constants require the effective nonlinear optical coefficient $d_{\rm eff}$ for the type I phase-matched interaction of SHG ($\theta = \theta_m$), κ_1 and the phase-mismatched interaction FHG(ee), κ_2. We determined $d_{\rm eff2}$ by the use of two e-polarized SH electric fields combining to produce an e-polarized FH electric field. In $LiNbO_3$ these are

$$d_{\rm eff1} = d_{31} \sin\theta - d_{22} \cos\theta \sin 3\Phi, \tag{2a}$$

$$d_{\rm eff2} = d_{31} \cos^2\theta \sin\theta + d_{22} \cos^3\theta \sin 3\Phi + d_{33} \sin^3\theta. \tag{2b}$$

The $LiNbO_3$ crystal is cut such that $\Phi = 90°$. FHG coupling, κ_2, takes advantage of the large value of d_{33} = 33 pm/V (at 1.06 μm), and at 2 μm, $\kappa_2 = 42.8 \times 10^{-6}\,V^{-1}$.

A standard numerical routine is used to solve the coupled equations.[11] This routine uses the Gear method, which is best suited for solving a difficult problem, such as these coupled nonlinear differential equations. The integration step size Δz was chosen such that $\Delta k_{4\omega}\Delta z = 0.01$ so that the oscillation of the fields could be accurately followed. Conservation of energy was checked for each step through the crystal. Equations (1) can also be solved analytically for our experimental conditions. Because the conversion efficiency is so high for the first-stage SHG (>50%), pump-depletion effects are important, and the solution for the SH field is $E_2(z) = E_1(0) \tanh[\kappa_1 |E_1(0)| z]$. We obtain an approximate solution for the FH electric field, $|E_4(z)| = \kappa_2 (E_2)^2/\Delta k_{4\omega}\, |\exp(-i\Delta k_{4\omega} z) - 1|$ by integrating Eq. (1c) with the above form for $E_2(z)$, where $E_2(z)$ is slowly varying compared with the fast-oscillating exponential term and is taken outside the integral. This approximate solution to Eqs. (1) is appropriate because the first-stage SHG is phase matched, $\Delta k_{2\omega} = 0$; the phase mismatch to the FH is large, $\Delta k_{4\omega}$ = 12,150 rad/cm; and the FH field is weak. Figure 2(a) shows the results of the approximate analytical solution and the numerical simula-

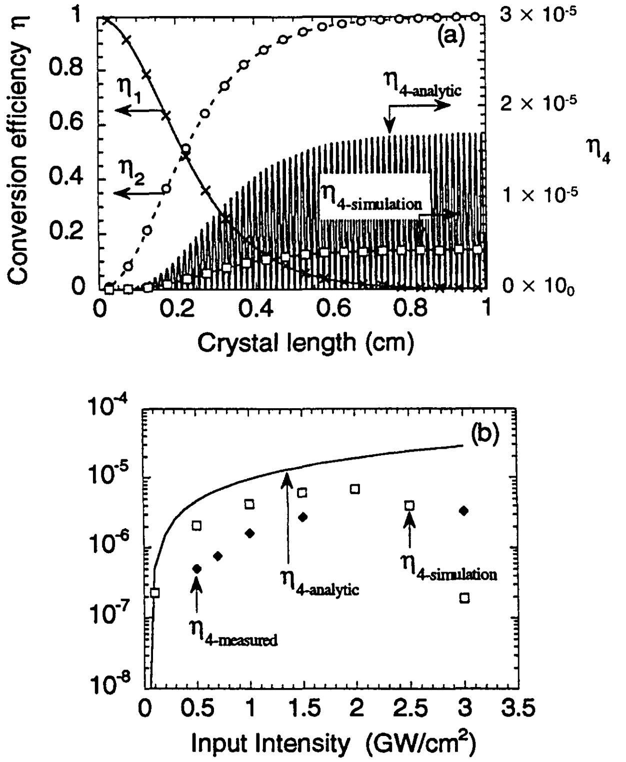

Fig. 2. (a) Energy conversion efficiencies as a function of crystal length for the fundamental, $\eta_1 = \mathscr{E}_1/\mathscr{E}_1$, the SH, $\eta_2 = \mathscr{E}_2/\mathscr{E}_1$, and the FH, $\eta_4 = \mathscr{E}_4/\mathscr{E}_1$ from an approximate analytical solution (solid curve with x, η_1; dashed curve with ○, η_2; solid curve, η_4) and a numerical simulation (χ, η_1, ○, η_2; □, η_4) of Eqs. (1) in $LiNbO_3$ at 2 μm for the following parameters: $\Delta k_{2\omega} = 0$ rad/cm, $\Delta k_{4\omega} =$ 12,150 rad/cm, $\kappa_1 = 6.8 \times 10^{-6}$ ($\Phi = 90°$), $\kappa_2 = 42.8 \times 10^{-6}$ and $I_1(0) = 1.0$ GW/cm^2. (b) Comparison of the experimentally measured FHG (♦, $\eta_{4\text{-measured}}$), the numerical simulation of FHG (□, $\eta_{4\text{-simulation}}$), and the analytically calculated (solid curve, $\eta_{4\text{-analytic}}$) conversion efficiencies for the FH versus input intensity in 1 cm of $LiNbO_3$ at 2 μm.

tion for energy ($\mathscr{E}$) conversion efficiencies of the fundamental, $\eta_1 = \mathscr{E}_1/\mathscr{E}_1$, the SH, $\eta_2 = \mathscr{E}_2/\mathscr{E}_1$, and the FH, $\eta_4 = \mathscr{E}_4/\mathscr{E}_1$, as a function of crystal length for the following input parameters: $\Delta k_{2\omega} = 0$ rad/cm, $\Delta k_{4\omega} = 12{,}150$ rad/cm, $\kappa_1 = 6.8 \times 10^{-6}\,\mathrm{V}^{-1}$ ($\Phi = 90°$), $\kappa_2 = 42.8 \times 10^{-6}\,\mathrm{V}^{-1}$, and $I_1(0) = 1.0$ GW/cm^2 for $LiNbO_3$ at 2 μm. The FH conversion efficiency, for the above parameters in 1 cm of crystal is $\eta_{4\text{-analytic}} = 9.4 \times 10^{-6}$ (analytical approximation) and $\eta_{4\text{-simulation}} = 4.2 \times 10^{-6}$ (numerical simulation), compared with the measured value $\eta_{4\text{-measured}} = 1.6 \times 10^{-6}$. Note the close agreement between the numerical simulation and the analytical approximation for the fundamental and the SH conversion efficiencies and the failure of the approximation for the FH. The dispersion data of Edwards and Lawrence for $LiNbO_3$ were used for these calculations.[12]

The FH conversion efficiency for the analytical approximation, $\eta_{4\text{-analytic}}$, exhibits fast oscillations because of the large phase mismatch, $\Delta k_{4\omega} = 12{,}150$ rad/cm. In our experiment, however, the phase mismatch is not a discrete number. Both the continuous distribution in wavelengths, from the 0.5% FWHM wavelength spectrum, and the increase in intensity through the pulse are responsible for a continuous phase mismatch $\Delta k_{4\omega}$ through the crystal. Thus we are only concerned with the amplitude envelope of $\eta_{4\text{-analytic}}$ and not with the fast oscillations. The envelope of $\eta_{4\text{-analytic}}$ therefore gives us a conversion efficiency of 1.7×10^{-5}, 4 times the simulation value and an order of magnitude larger than the measured value. In Fig. 2(b) we compare experimentally measured FHG's (♦), numerical simulation of FHG (□), and analytically calculated (solid line) conversion efficiencies for the FH versus input intensity in $LiNbO_3$ at 2 μm. Note that as the input intensity increases the simulation curve for FHG, $\eta_{4\text{-simulation}}$, starts to decrease for a 1-cm-long crystal; the $\eta_{4\text{-simulation}}$ curve narrows as the intensity increases. This behavior was not observed experimentally.

At 2.5 μm the simulated conversion efficiencies for FHG are approximately an order of magnitude better than at 2 μm. However, the measured conversion efficiencies for FHG at 2 and 2.5 μm were approximately the same order of magnitude and slightly smaller than those calculated for 2 μm. For the simulations we assume a monochromatic plane wave and do not take into account the group velocity walkoff of these short (2-ps) pulses. For propagation in 1 cm of $LiNbO_3$ crystal the group velocity walkoff is 64 fs at 2 μm for the fundamental and 1.2 ps at 2.5 μm for the SH. The large group velocity walkoff could be responsible for the lower measured conversion efficiency at 2.5 μm. The crystal is antireflection coated on the exit face down to a wavelength of 0.8 μm. The FH in these experiments (0.5–0.625 μm) could have been experiencing some loss on exiting the crystal at these shorter wavelengths. No third harmonic was observed in these experiments.

An analysis of third-harmonic generation (THG), from a mixing of the fundamental and the generated SH, shows that the coupling for THG is approximately an order of magnitude less than the coupling for FHG. And the phase mismatch for this THG interaction is of the same magnitude as for the FHG interaction. The numerically simulated THG conversion efficiency of $\eta_3 \sim 10^{-10}$ is 4 orders of magnitude smaller than that for FHG, $\eta_4 \sim 10^{-6}$, and so it is not surprising that we did not observe THG.

Conclusions

In summary, we have observed SHG and FHG in a single 1-cm-long $LiNbO_3$ crystal, type I phase matched for the first-stage SHG and pumped by 2–2.5-μm photons from the Mark III FEL. Energy conversion efficiencies as high as 66% for SHG and 3.3×10^{-6} for FHG were measured. The FH can be observed when the input energy per macropulse is at least 33 μJ (100 MW/cm^2). An analysis of coupled Eqs. (1) reveals that the FH was generated by the SH in a cascaded interaction, SHG:SHG, with a conversion efficiency of $\eta_4 \sim 10^{-6}$. The polarization of this cascaded interaction, type I (o ray + o ray → e ray) for the first stage and (e ray + e ray → e ray) for the second stage, was necessary for determining the coupling constants κ_1 and κ_2 and the phase mismatches $\Delta k_{2\omega}$ and $\Delta k_{4\omega}$. Calculated conversion efficiencies for FHG at 2.5 μm are approximately an order of magnitude larger than at 2 μm, but measured conversion efficiencies were similar over the range of 2 to 2.5 μm. The simulations do not account for group velocity walkoff of the pulses, and the exit face of the crystal is not antireflection coated for the FH. Group velocity walkoff and the lack of an antireflection coating could contribute to the lower measured FHG conversion efficiencies. No third harmonic was observed, but an analysis of THG, from a mixing of the fundamental and the generated SH, shows that the coupling for THG is approximately an order of magnitude less than the coupling for FHG and that the phase mismatch for THG is of the same magnitude as that for FHG. The THG conversion efficiency of $\eta_3 \sim 10^{-10}$ is 4 orders of magnitude smaller than that for FHG, and so it is not surprising that we did not observe THG. An approximate analytical solution to the coupled Eqs. (1) was developed to compare with the numerical simulation and experimental results, and the comparison showed reasonable agreement.

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